Practical Exam Guide

Dot Net Core and Progressive Web Application | T.Y.B.Sc. I.T.
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Is guide mein 61 slips mein se har slip ka sabse EASY question chuna gaya hai. Pass hone ke liye sirf 1 question pura karna hai. Neeche table mein apni slip number dhundho, uska Code number dekho (A1, B1, C1 ...), phir us code ka section kholo: usme project banane se lekar run karne tak ke saare steps aur poora code diya hai.

Tools: Visual Studio 2026. Web Forms (Part D) sirf full Visual Studio mein chalta hai.

Slip Lookup Table (Slip -> Question -> Code)

Slip	Do	Code	Slip	Do	Code
1	Q1	A1	32	Q1	A7
2	Q1	A2	33	Q2	A6
3	Q1	A3	34	Q1	A3
4	Q1	A4	35	Q1	D2
5	Q1	A5	36	Q2	A6
6	Q2	B1	37	Q2	A2
7	Q1	A6	38	Q2	C1
8	Q1	A7	39	Q1	A10
9	Q2	C1+C3	40	Q1	A4
10	Q2	C4	41	Q1	A3
11	Q2	C1+C2	42	Q1	A2
12	Q2	B1	43	Q1	A8
13	Q2	C1+C2	44	Q1	A5
14	Q1	A7	45	Q1	A1
15	Q2	C1	46	Q2	C1+C2
16	Q1	A8	47	Q2	C1+C2
17	Q2	A6	48	Q2	C1+C3
18	Q1	A7	49	Q2	C4
19	Q1	A8	50	Q2	C1+C3
20	Q1	A9	51	Q2	C1
21	Q1	A3	52	Q2	C1
22	Q1	D1	53	Q2	C1+C2
23	Q1	A8	54	Q1	A4
24	Q2	A8	55	Q1	A4
25	Q1	D2	56	Q2	B1
26	Q1	C1+C2	57	Q2	C4
27	Q1	D3	58	Q2	B1
28	Q1	C1	59	Q2	B1
29	Q1	C1	60	Q1	A6+A7
30	Q2	A8	61	Q2	C1+C2
31	Q2	A7			

Note: Slip 22, 25, 26, 27, 28, 29, 35 mein original PDF mein sirf Q1 mila tha. Paper mein Q2 bhi ho toh dekh lena. Slip 60 mein A6 aur A7 dono (Floyd + Cities) karne hain. Code 'C1+C2' ka matlab: pehle C1 banao, phir C2 ke changes karo.

Project Type Cheat Sheet (new project banate waqt kya select karna hai)

Code	Visual Studio mein ye template chuno	Additional information screen
A1 - A10	Console App (C#)	Framework: latest .NET. Top-level statements wala option untick.
B1	ASP.NET Core Web API	Authentication: None. 'Use controllers' TICK. Container support untick.
C1, C2, C3, C4	ASP.NET Core Empty (ya teacher bole toh ASP.NET Core Web App (Model-View-Controller))	Framework: latest .NET. HTTPS tick. Container support untick. MVC mein Authentication: None.
D1, D2, D3	ASP.NET Web Application (.NET Framework) -> Web Forms	Framework: .NET Framework 4.8. Authentication: No Authentication.

Har project banate waqt: Visual Studio -> Create a new project -> upar search box mein template ka naam likho -> list se select -> Next -> Project name -> Next -> Additional information -> Create. Har code ke section mein neeche yellow box mein exact template likha hai.

PART A: C# Console Programs

Har program ke steps same hain. Sirf Project Name aur code alag hai.

A1: Delegates se Add / Subtract / Multiply / Divide

Slips: 1, 45 | Project Name: DelegateDemo | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: DelegateDemo (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace DelegateDemo
{
    public delegate void Calculation(int a, int b);

    class Arithmetic
    {
        public void Add(int a, int b) { Console.WriteLine("Addition = " + (a + b)); }
        public void Subtract(int a, int b) { Console.WriteLine("Subtraction = " + (a - b)); }
        public void Multiply(int a, int b) { Console.WriteLine("Multiplication = " + (a * b)); }
        public void Divide(int a, int b) { Console.WriteLine("Division = " + (a / b)); }
    }

    class Program
    {
        static void Main()
        {
            Arithmetic obj = new Arithmetic();
            Calculation c;
            c = obj.Add; c(20, 10);
            c = obj.Subtract; c(20, 10);
            c = obj.Multiply; c(20, 10);
            c = obj.Divide; c(20, 10);
            Console.ReadLine();
        }
    }
}
```

A2: Function Overloading + Constructor Overloading

Slips: 2, 37, 42 | Project Name: OverloadingDemo | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: OverloadingDemo (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace OverloadingDemo
{
    class Calculator
    {
        public int Add(int a, int b) { Console.WriteLine("Add(int,int) called"); return a + b; }
        public int Add(int a, int b, int c) { Console.WriteLine("Add(int,int,int) called"); return a + b +
            c; }
        public double Add(double a, double b) { Console.WriteLine("Add(double,double) called"); return a +
            b; }
        public string Add(string a, string b) { Console.WriteLine("Add(string,string) called"); return a +
            b; }
    }

    class Student
    {
        public Student() { Console.WriteLine("Default Constructor Called"); }
        public Student(string name) { Console.WriteLine("Student Name: " + name); }
        public Student(string name, int age)
        {
            Console.WriteLine("Student Name: " + name);
            Console.WriteLine("Student Age: " + age);
        }
    }

    class Program
    {
        static void Main()
        {
            Console.WriteLine("=== Function Overloading ===");
            Calculator calc = new Calculator();
            Console.WriteLine(calc.Add(10, 20));
            Console.WriteLine(calc.Add(10, 20, 30));
            Console.WriteLine(calc.Add(1.5, 2.5));
            Console.WriteLine(calc.Add("Hello, ", "World!"));

            Console.WriteLine("\n=== Constructor Overloading ===");
            Student s1 = new Student();
            Student s2 = new Student("Ganesh");
            Student s3 = new Student("Raj", 19);
            Console.ReadLine();
        }
    }
}
```

A3: Interface: Area of Circle and Rectangle (IShape)

Slips: 3, 21, 34, 41 | Project Name: InterfaceDemo | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: InterfaceDemo (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace InterfaceDemo
{
    interface IShape
    {
        double CalculateArea();
    }

    class Circle : IShape
    {
        double radius;
        public Circle(double r) { radius = r; }
        public double CalculateArea() { return Math.PI * radius * radius; }
    }

    class Rectangle : IShape
    {
        double length, breadth;
        public Rectangle(double l, double b) { length = l; breadth = b; }
        public double CalculateArea() { return length * breadth; }
    }

    class Program
    {
        static void Main()
        {
            IShape s1 = new Circle(5);
            IShape s2 = new Rectangle(10, 4);
            Console.WriteLine("Area of Circle = " + s1.CalculateArea());
            Console.WriteLine("Area of Rectangle = " + s2.CalculateArea());
            Console.ReadLine();
        }
    }
}
```

A4: Delegates + Multicast Delegate + Events

Slips: 4, 40, 54, 55 | Project Name: DelegateEventDemo | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: DelegateEventDemo (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace DelegateEventDemo
{
    public delegate void Demo();
    public delegate void NotifyHandler(string message);

    class Test
    {
        public void Show() { Console.WriteLine("show method called"); }
        public void Display() { Console.WriteLine("display method called"); }
        public void Print() { Console.WriteLine("print method called"); }
    }

    class EventPublisher
    {
        public event NotifyHandler OnDataProcessed;
        public void ProcessData(string data)
        {
            Console.WriteLine("Processing data: " + data);
            if (OnDataProcessed != null)
                OnDataProcessed("Data '" + data + "' has been processed!");
        }
    }

    class EventSubscriber
    {
        public void HandleEvent(string message)
        {
            Console.WriteLine("[Subscriber] Received: " + message);
        }
    }

    class Program
    {
        static void Main()
        {
            Console.WriteLine("=== Multicast Delegate ===");
            Test t = new Test();
            Demo d;
            d = t.Show;
            d += t.Display;
            d += t.Print;
            d();

            Console.WriteLine("\n=== Event Demo ===");
            EventPublisher publisher = new EventPublisher();
            EventSubscriber subscriber = new EventSubscriber();
            publisher.OnDataProcessed += subscriber.HandleEvent;
            publisher.ProcessData("Student Record");
            publisher.ProcessData("Salary Data");
            Console.ReadLine();
        }
    }
}
```

A5: BankAccount (balance negative nahi ho sakta)

Slips: 5, 44 | Project Name: BankDemo | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: BankDemo (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace BankDemo
{
    class BankAccount
    {
        private double balance;
        public int AccountNumber { get; set; }

        public double Balance
        {
            get { return balance; }
            set
            {
                if (value >= 0)
                    balance = value;
                else
                    Console.WriteLine("Balance cannot be negative");
            }
        }
    }

    class Program
    {
        static void Main()
        {
            BankAccount b = new BankAccount();
            b.AccountNumber = 101;
            b.Balance = 5000;
            Console.WriteLine("Account Number: " + b.AccountNumber);
            Console.WriteLine("Balance: " + b.Balance);
            b.Balance = -2000;
            Console.WriteLine("Balance after invalid update: " + b.Balance);
            Console.ReadLine();
        }
    }
}
```

A6: Floyd's Triangle (n rows user se)

Slips: 7, 17, 33, 36, 60 | Project Name: FloydTriangle | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: FloydTriangle (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace FloydTriangle
{
    class Program
    {
        static void Main()
        {
            int n, num = 1;
            Console.Write("Enter the number of rows: ");
            n = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("\nFloyd's Triangle:\n");
            for (int i = 1; i <= n; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }
            Console.ReadLine();
        }
    }
}
```

A7: 10 Cities / Employees ko alphabetical order mein display karo

Slips: 8, 14, 18, 31, 32, 60 | Project Name: CitySorting | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: CitySorting (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace CitySorting
{
    class Program
    {
        static void Main()
        {
            string[] cities = new string[10];
            Console.WriteLine("Enter names of 10 cities:");
            for (int i = 0; i < 10; i++)
            {
                Console.Write("City " + (i + 1) + ": ");
                cities[i] = Console.ReadLine();
            }

            Array.Sort(cities);

            Console.WriteLine("\nCities in alphabetical order:");
            foreach (string c in cities)
            {
                Console.WriteLine(c);
            }
            Console.ReadLine();
        }
    }
}
```

Slip 14 (Employees): code mein 'cities' ki jagah 'employees' aur 'City' ki jagah 'Employee' likh do. Baaki sab same.

A8: Inheritance: Single, Multilevel, Hierarchical, Multiple (interface)

Slips: 16, 19, 23, 24, 30, 43 | Project Name: InheritanceDemo | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: InheritanceDemo (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace InheritanceDemo
{
    // Single inheritance
    class Animal { public void Eat() { Console.WriteLine("Animal is eating"); } }
    class Dog : Animal { public void Bark() { Console.WriteLine("Dog is barking"); } }

    // Multilevel inheritance
    class GrandParent { public void Method1() { Console.WriteLine("GrandParent method"); } }
    class Parent : GrandParent { public void Method2() { Console.WriteLine("Parent method"); } }
    class Child : Parent { public void Method3() { Console.WriteLine("Child method"); } }

    // Hierarchical inheritance
    class Shape { public void Draw() { Console.WriteLine("Drawing a shape"); } }
    class Circle : Shape { public void Info() { Console.WriteLine("This is a circle"); } }
    class Rectangle : Shape { public void Info() { Console.WriteLine("This is a rectangle"); } }

    // Multiple inheritance (using interfaces)
    interface IFlyable { void Fly(); }
    interface ISwimmable { void Swim(); }
    class Duck : IFlyable, ISwimmable
    {
        public void Fly() { Console.WriteLine("Duck can fly"); }
        public void Swim() { Console.WriteLine("Duck can swim"); }
    }

    class Program
    {
        static void Main()
        {
            Console.WriteLine("--- Single Inheritance ---");
            Dog d = new Dog();
            d.Eat(); d.Bark();

            Console.WriteLine("--- Multilevel Inheritance ---");
            Child c = new Child();
            c.Method1(); c.Method2(); c.Method3();

            Console.WriteLine("--- Hierarchical Inheritance ---");
            Circle cir = new Circle(); cir.Draw(); cir.Info();
            Rectangle rec = new Rectangle(); rec.Draw(); rec.Info();

            Console.WriteLine("--- Multiple Inheritance (interface) ---");
            Duck duck = new Duck();
            duck.Fly(); duck.Swim();
            Console.ReadLine();
        }
    }
}
```

Poora program run kar do. Slip mein 'single' mange, 'multilevel' mange ya 'any two types', sab output mein aa jayega.

A9: Multiple Inheritance using IPrint and IShow (class Demo)

Slips: 20 | Project Name: MultipleInheritance | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: MultipleInheritance (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace MultipleInheritance
{
    interface IPrint { void Print(); }
    interface IShow { void Show(); }

    class Demo : IPrint, IShow
    {
        public void Print() { Console.WriteLine("Print method of IPrint"); }
        public void Show() { Console.WriteLine("Show method of IShow"); }
    }

    class Program
    {
        static void Main()
        {
            Demo d = new Demo();
            d.Print();
            d.Show();
            Console.ReadLine();
        }
    }
}
```

A10: Exception Handling (try, catch, finally, divide by zero)

Slips: 39 | Project Name: ExceptionHandling | Project Type: Console App (C#)

TEMPLATE SELECT KARO: Console App (C#) | Language: C# | Platform: All platforms | Project type: Console | Framework: latest .NET (default)

Step 1: Visual Studio open karo -> start window mein 'Create a new project' par click karo.

Step 2: Upar search box mein 'Console App' likho. Uske side ke filters: Language = C#, Platform = All platforms, Project type = Console. List mein 'Console App' (C# icon wala) select karo. 'Console App (.NET Framework)' ya 'Windows Forms' mat chuno -> Next dabao.

Step 3: Project name mein likho: ExceptionHandling (Location jaisi hai waisi rehne do) -> Next dabao.

Step 4: 'Additional information' screen aayegi: Framework = latest .NET (default) rehne do. 'Do not use top-level statements' ko UNTICK hi rehne do. Koi aur option change mat karo -> Create dabao.

Step 5: Right side Solution Explorer mein 'Program.cs' par double-click karo (code editor mein khulega).

Step 6: Editor mein click karo -> Ctrl + A (sab select) -> Delete dabao -> neeche diya hua code copy karke Ctrl + V se paste karo.

Step 7: Ctrl + S se save karo -> Ctrl + F5 dabao (Start Without Debugging). Output console window mein aayega.

Step 8: Jahan input maange (numbers / names) wahan type karke Enter dabao.

Program.cs (poora code)

```
using System;
namespace ExceptionHandling
{
    class Program
    {
        static void Main()
        {
            int a, b, result;
            try
            {
                Console.Write("Enter First Number: ");
                a = Convert.ToInt32(Console.ReadLine());
                Console.Write("Enter Second Number: ");
                b = Convert.ToInt32(Console.ReadLine());
                result = a / b;
                Console.WriteLine("Result = " + result);
            }
            catch (DivideByZeroException)
            {
                Console.WriteLine("Cannot divide by zero");
            }
            catch (FormatException)
            {
                Console.WriteLine("Invalid Input");
            }
            finally
            {
                Console.WriteLine("Program Ended");
            }
            Console.ReadLine();
        }
    }
}
```

Test: dusra number 0 daalo, 'Cannot divide by zero' aayega, phir finally ka message bhi aayega.

PART B: ASP.NET Core Web API

B1: Student JSON API + Factorial + Digit Sum

Slips: 6, 56, 59 (Student JSON) | 12 (Digit sum) | 58 (Factorial) | Project Name: StudentAPI | Type: ASP.NET Core Web API

TEMPLATE SELECT KARO: ASP.NET Core Web API | Language: C# | Platform: All platforms | Project type: Web (API) | Additional info: Framework latest .NET, Authentication type = None, Configure for HTTPS = tick, 'Use controllers' = TICK (zaroori), Enable OpenAPI = tick ya untick dono chalega, Enable container support = UNTICK

Step 1: Visual Studio open karo -> 'Create a new project' par click karo.

Step 2: Search box mein 'ASP.NET Core Web API' likho. Filters: Language = C#, Platform = All platforms, Project type = Web ya API. List mein 'ASP.NET Core Web API' (sirf ye, 'minimal' ya 'native AOT' wala nahi) select karo -> Next dabao.

Step 3: Project name: StudentAPI -> Next dabao.

Step 4: 'Additional information' screen: Framework = latest .NET, Authentication type = None, Configure for HTTPS = tick, 'Use controllers' = TICK karo, Enable container support = untick -> Create dabao.

Step 5: Solution Explorer mein 'WeatherForecast.cs' par right-click -> Delete -> OK.

Step 6: Solution Explorer mein 'Controllers' folder kholo -> 'WeatherForecastController.cs' par right-click -> Delete -> OK.

Step 7: Project name 'StudentAPI' par right-click -> Add -> New Item -> 'Class' select karo -> Name mein 'MyApiController.cs' likho -> Add dabao.

Step 8: Is file ka poora code delete karke neeche wala 'MyApiController.cs' code paste karo.

Step 9: Solution Explorer mein 'Program.cs' kholo -> poora code delete karo -> neeche wala Program.cs code paste karo.

Step 10: Ctrl + S -> Ctrl + F5 dabao. Browser khulega (kabhi 404 ya Swagger page dikhe toh tension nahi).

Step 11: Browser ki address bar mein sirf URL ka end badlo (port number jo dikh raha hai wahi rakho):

Slip 6, 56, 59: /api/students | Slip 12: /api/digitsum/1234 (answer 10) | Slip 58: /api/factorial/5 (answer 120)

Postman maange toh: Postman kholo -> method GET -> poora URL likho, jaise http://localhost:PORT/api/factorial/5 -> Send dabao. Neeche JSON response dikhega.

MyApiController.cs (poora code)

```
using Microsoft.AspNetCore.Mvc;

namespace StudentAPI.Controllers
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; } = "";
        public string Course { get; set; } = "";
        public int Marks { get; set; }
    }

    [ApiController]
    [Route("api")]
    public class MyApiController : ControllerBase
    {
        [HttpGet("students")]
        public List<Student> GetStudents()
        {
            return new List<Student>()
            {
                new Student { RollNo = 1, Name = "Rahul", Course = "BSc IT", Marks = 85 },
                new Student { RollNo = 2, Name = "Sneha", Course = "BCA", Marks = 78 },
                new Student { RollNo = 3, Name = "Amit", Course = "BSc CS", Marks = 90 }
            };
        }

        [HttpGet("factorial/{n}")]
        public IActionResult Factorial(int n)
        {
            if (n < 0) return BadRequest("Negative number not allowed");
            long f = 1;
            for (int i = 1; i <= n; i++) f *= i;
            return Ok(new { number = n, factorial = f });
        }

        [HttpGet("digitsum/{n}")]
        public IActionResult DigitSum(int n)
        {
            int sum = 0;
            int t = Math.Abs(n);
            while (t > 0)
            {
                sum += t % 10;
                t /= 10;
            }
            return Ok(new { number = n, digitSum = sum });
        }
    }
}
```

Program.cs (poora code)

```
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllers();
var app = builder.Build();
app.MapControllers();
app.Run();
```

PART C: PWA (Progressive Web Application)

C1: Base PWA (Manifest + Service Worker + Offline Page + Cache)

Slips: 15, 28, 29, 38, 51, 52 (sirf C1) | 9, 48, 50 (C1+C3) | 11, 13, 26, 46, 47, 53, 61 (C1+C2) | 10, 49, 57 (C4) | Project Name: MyPWA | Type: ASP.NET Core Empty

TEMPLATE SELECT KARO: ASP.NET Core Empty | Language: C# | Platform: All platforms | Project type: Web | Additional info: Framework latest .NET, Configure for HTTPS = tick, Enable container support = UNTICK. (College Practical 10 mein bhi yahi 'ASP.NET Core Empty Web App' use hua hai.)

Agar teacher MVC bole (College Practical 7 aur 8 mein MVC use hua tha): template 'ASP.NET Core Web App (Model-View-Controller)' select karo, Authentication type = None. Us case mein 'wwwroot' folder pehle se bana hota hai, isliye uska New Folder wala step skip karo. Baaki saari files (index.html, manifest.json, service-worker.js, app.js, offline.html) wwwroot mein waise hi banao aur Program.cs bhi neeche wala hi paste karo (College Practical 8 mein bhi MVC project mein yehi Program.cs use hua hai).

Step 1: Visual Studio open karo -> 'Create a new project'.

Step 2: Search box mein 'ASP.NET Core Empty' likho. Filters: Language = C#, Platform = All platforms, Project type = Web. List mein 'ASP.NET Core Empty' select karo -> Next.

Step 3: Project name: MyPWA -> Next -> 'Additional information' screen: Framework = latest .NET, Configure for HTTPS = tick, Enable container support = untick -> Create dabao.

Step 4: Solution Explorer mein 'Program.cs' kholo -> poora code delete karke neeche wala Program.cs code paste karo.

Step 5: Project name 'MyPWA' par right-click -> Add -> New Folder -> naam do: wwwroot (bilkul yahi spelling).

Step 6: 'wwwroot' par right-click -> Add -> New Folder -> naam do: icons.

Step 7: Koi bhi 2 PNG images lo (Google se ya Paint se bana lo). File Explorer se copy karke 'icons' folder mein Visual Studio mein paste karo aur rename karo: icon1.png aur icon2.png.

Step 8: 'wwwroot' par right-click -> Add -> New Item -> 'HTML Page' select -> Name: index.html -> Add. Purana code delete karke neeche wala index.html paste karo.

Step 9: 'wwwroot' par right-click -> Add -> New Item -> 'JSON File' (agar na mile toh 'Text File') -> Name: manifest.json -> Add. Code paste karo. (Text File ho toh naam badal ke .json hi rakho.)

Step 10: 'wwwroot' par right-click -> Add -> New Item -> 'JavaScript File' -> Name: service-worker.js -> Add. Code paste karo.

Step 11: 'wwwroot' par right-click -> Add -> New Item -> 'JavaScript File' -> Name: app.js -> Add. Code paste karo.

Step 12: 'wwwroot' par right-click -> Add -> New Item -> 'HTML Page' -> Name: offline.html -> Add. Purana code delete karke code paste karo.

Step 13: Sab files Ctrl + S se save karo. Ctrl + F5 dabao. Chrome mein page khulega. Ek baar F5 se refresh karo.

Step 14: Dikhane ke liye: F12 dabao -> upar 'Application' tab -> left side 'Manifest' (name, icons, start URL, theme color dikhega) -> 'Service Workers' (status: activated and running).

Step 15: Offline test: F12 mein 'Network' tab -> 'Offline' checkbox tick karo -> F5 dabao -> page phir bhi chalega.

Step 16: Custom offline page dikhane ke liye: Offline hote hue address bar mein localhost:PORT/abcd likho -> 'You are offline' wala offline.html page dikhega.

Slip 15 mein sirf HTML page + manifest.json chahiye: yahi project banao, F12 -> Application -> Manifest dikha do.

Program.cs

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

wwwroot/index.html

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>My PWA</title>
    <link rel="manifest" href="manifest.json">
    <meta name="theme-color" content="#0d6efd">
</head>
<body>
    <h1>My Progressive Web App</h1>
    <p>Ye application internet ke bina bhi chalegi.</p>
    <script src="app.js"></script>
</body>
</html>
```

wwwroot/manifest.json

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    { "src": "icons/icon1.png", "sizes": "192x192", "type": "image/png" },
    { "src": "icons/icon2.png", "sizes": "512x512", "type": "image/png" }
  ]
}
```

wwwroot/service-worker.js (cache-first strategy)

```
const CACHE_NAME = "pwa-cache-v1";
const urlsToCache = ["/", "/index.html", "/offline.html", "/app.js", "/manifest.json"];

self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME).then(cache => cache.addAll(urlsToCache))
    );
    self.skipWaiting();
});

self.addEventListener("activate", event => {
    event.waitUntil(self.clients.claim());
});

self.addEventListener("fetch", event => {
    event.respondWith(
        caches.match(event.request).then(response => {
            return response || fetch(event.request).catch(() => caches.match("/offline.html"));
        })
    );
});
```

wwwroot/app.js

```
if ("serviceWorker" in navigator) {
    navigator.serviceWorker.register("/service-worker.js")
        .then(() => console.log("Service Worker Registered"))
        .catch(err => console.error("Registration failed", err));
}
```

wwwroot/offline.html

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <title>Offline</title>
</head>
<body style="text-align:center; font-family:Arial; margin-top:80px;">
    <h1>You are offline</h1>
    <p>Internet connection nahi hai. Connection aane ke baad dobara try karo.</p>
</body>
</html>
```

C2: Website Content (College / Hotel / Company) on top of C1

Slips: 11 (Hotel), 13 (Company), 26, 46, 47, 53, 61 (College)

Step 1: Pehle C1 ke saare steps pure karo (project MyPWA banao).

Step 2: Solution Explorer mein wwwroot -> 'index.html' par double-click karo.

Step 3: <body> aur </body> ke andar ka poora content delete karo aur neeche wala College body paste karo (<script src="app.js"> line rakhna).

Step 4: Hotel (Slip 11): headings About, Rooms, Food, Contact rakho aur content hotel ke hisaab se likho. Company (Slip 13): headings About Us, Product, Services, Contact.

Step 5: Ctrl + S -> Ctrl + F5 -> F5 refresh -> F12 -> Application tab mein Manifest aur Service Worker dikhao -> Network -> Offline -> F5.

index.html ka body (College)

```
<body>
    <h1>ABC College of Science</h1>
    <p>
        <a href="#about">About</a> |
        <a href="#courses">Courses</a> |
        <a href="#faculty">Faculty</a> |
        <a href="#contact">Contact</a>
    </p>

    <h2 id="about">About</h2>
    <p>Our college provides quality education in science and technology.</p>

    <h2 id="courses">Courses</h2>
    <ul>
        <li>BSc IT</li>
        <li>BSc CS</li>
        <li>BCA</li>
    </ul>

    <h2 id="faculty">Faculty</h2>
    <ul>
        <li>Prof. A. Sharma - Programming</li>
        <li>Prof. S. Patil - Database</li>
        <li>Prof. R. Khan - Networking</li>
    </ul>

    <h2 id="contact">Contact</h2>
    <p>Email: info@abccollege.edu | Phone: 022-12345678</p>

    <script src="app.js"></script>
</body>
```

C3: Push Notification on top of C1

Slips: 9, 48, 50

Step 1: Pehle C1 ke saare steps pure karo.

Step 2: wwwroot -> index.html kholo. '<script src="app.js"></script>' line se theek pehle neeche wali button line add karo.

Step 3: wwwroot -> app.js kholo. Purane code ke NEECHE (end mein) showNotification function paste karo. Purana code delete mat karna.

Step 4: Ctrl + S -> Ctrl + F5. Page par 'Show Notification' button dabao.

Step 5: Browser permission maangega -> 'Allow' dabao -> Notification aa jayega (screen ke corner mein).

index.html mein add karo

```
<button onclick="showNotification()">Show Notification</button>
```

app.js ke end mein add karo

```
function showNotification() {
    Notification.requestPermission().then(permission => {
        if (permission === "granted") {
            navigator.serviceWorker.ready.then(registration => {
                registration.showNotification("PWA Notification", {
                    body: "Push Notification Working Successfully!",
                    icon: "icons/icon1.png"
                });
            });
        }
    });
}
```

C4: E-Commerce PWA with Product List and Offline Cart

Slips: 10, 49, 57

Step 1: Pehle C1 ke saare steps pure karo (MyPWA project, manifest, service-worker.js, offline.html waise hi rahenge).

Step 2: wwwroot -> index.html kholo -> poora code delete -> neeche wala C4 index.html paste.

Step 3: wwwroot -> app.js kholo -> poora code delete -> neeche wala C4 app.js paste.

Step 4: Ctrl + S -> Ctrl + F5. Products ke 'Add to Cart' button dabao, left side cart mein items dikhenge.

Step 5: Page refresh karo: cart waise ka waisa rahega (localStorage).

Step 6: F12 -> Network -> Offline tick -> F5: page aur cart offline mein bhi chalte hain.

Step 7: Agar purana page dikhe toh F12 -> Application -> Storage -> 'Clear site data' dabao -> F5.

wwwroot/index.html (C4)

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>E-Commerce PWA</title>
    <link rel="manifest" href="manifest.json">
    <meta name="theme-color" content="#0d6efd">
    <style>
        body { font-family: Arial; margin: 20px; }
        .container { display: flex; gap: 30px; }
        .cart-section { width: 30%; border: 1px solid gray; padding: 15px; min-height: 300px; }
        .product-section { width: 70%; }
        .products { display: flex; flex-wrap: wrap; gap: 15px; }
        .product { border: 1px solid gray; padding: 10px; width: 200px; }
        button { padding: 5px 10px; cursor: pointer; }
    </style>
</head>
<body>
    <h1>E-Commerce PWA</h1>
    <div class="container">
        <div class="cart-section">
            <h2>Cart Items</h2>
            <ul id="cartList"></ul>
            <button onclick="clearCart()">Clear Cart</button>
        </div>
        <div class="product-section">
            <h2>Products</h2>
            <div class="products">
                <div class="product">
                    <h3>Laptop</h3>
                    <p>Price: Rs. 50000</p>
                    <button onclick="addToCart('Laptop')">Add to Cart</button>
                </div>
                <div class="product">
                    <h3>Mobile</h3>
                    <p>Price: Rs. 20000</p>
                    <button onclick="addToCart('Mobile')">Add to Cart</button>
                </div>
                <div class="product">
                    <h3>Headphones</h3>
                    <p>Price: Rs. 3000</p>
                    <button onclick="addToCart('Headphones')">Add to Cart</button>
                </div>
            </div>
        </div>
    </div>
    <script src="app.js"></script>
</body>
</html>
```

wwwroot/app.js (C4)

```
if ("serviceWorker" in navigator) {
    navigator.serviceWorker.register("/service-worker.js")
        .then(() => console.log("Service Worker Registered"));
}

let cart = JSON.parse(localStorage.getItem("cart")) || [];
displayCart();

function addToCart(product) {
    cart.push(product);
    localStorage.setItem("cart", JSON.stringify(cart));
    displayCart();
}

function clearCart() {
    cart = [];
    localStorage.setItem("cart", JSON.stringify(cart));
    displayCart();
}

function displayCart() {
    let cartList = document.getElementById("cartList");
    cartList.innerHTML = "";
    cart.forEach(item => {
        let li = document.createElement("li");
        li.textContent = item;
        cartList.appendChild(li);
    });
}
```

PART D: ASP.NET Web Forms (sirf full Visual Studio mein)

TEMPLATE SELECT KARO: ASP.NET Web Application (.NET Framework) -> next screen mein template: Web Forms | Language: C# | Framework: .NET Framework 4.8

Slips 22, 25, 27, 35 mein hi Web Forms aata hai. Pehle ye project setup karo (D1, D2, D3 teeno ke liye same):

Step 1: Visual Studio -> 'Create a new project'.

Step 2: Search box mein 'ASP.NET Web Application (.NET Framework)' likho. Filters: Language = C#, Platform = Windows, Project type = Web. List mein 'ASP.NET Web Application (.NET Framework)' C# wala select karo -> Next.

Step 3: Project name do (neeche har code mein likha hai) -> Location default rehne do -> Framework: '.NET Framework 4.8' -> Create dabao.

Step 4: Template list aayegi (Empty, Web Forms, MVC, Web API ...): isme 'Web Forms' select karo. MVC mat chuno. Authentication = No Authentication, Docker support = untick -> Create dabao.

Step 5: Agar ye template nahi mil raha toh Visual Studio Installer ka workload install karna padega (time lagta hai). Aisi halat mein teacher se poocho.

Page add karna: Solution Explorer mein project par right-click -> Add -> New Item -> 'Web Form' -> Name do -> Add. Page ke aspx file mein neeche 'Source' tab dabao. <form id="form1" runat="server"> ke andar jo khali   hai, use neeche ke markup se replace karo. Page par right-click -> 'Set As Start Page' -> Ctrl + F5.

D1: Student Registration Form (Label, TextBox, RadioButton, DropDownList, CheckBox, Calendar, Button)

Slip: 22 | Project Name: StudentRegistration | Page: StudentForm.aspx

Step 1: Upar wale setup se project 'StudentRegistration' banao.

Step 2: Project par right-click -> Add -> New Item -> Web Form -> Name: StudentForm.aspx -> Add.

Step 3: StudentForm.aspx ke 'Source' tab mein form tag ke andar wala replace karo (neeche markup).

Step 4: Solution Explorer mein StudentForm.aspx ke aage ka '>' arrow kholo -> 'StudentForm.aspx.cs' par double-click karo.

Step 5: Class ke curly brackets ke andar (Page_Load ke neeche) neeche wala btnSubmit_Click method paste karo.

Step 6: StudentForm.aspx par right-click -> Set As Start Page -> Ctrl + F5.

Step 7: Run hone par: naam likho, gender chuno, course chuno, calendar mein ek date par click karo, checkbox tick karo, Submit dabao. Output label mein dikhega.

StudentForm.aspx (div ki jagah)

```
<div>
    Student Name:
    <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
    <br /><br />
    Gender:
    <asp:RadioButton ID="rbMale" runat="server" GroupName="gender" Text="Male" />
    <asp:RadioButton ID="rbFemale" runat="server" GroupName="gender" Text="Female" />
    <br /><br />
    Course:
    <asp:DropDownList ID="ddlCourse" runat="server">
        <asp:ListItem>BSc IT</asp:ListItem>
        <asp:ListItem>BSc CS</asp:ListItem>
        <asp:ListItem>BCA</asp:ListItem>
    </asp:DropDownList>
    <br /><br />
    <asp:CheckBox ID="chkTerms" runat="server" Text="I accept Terms &amp; Conditions" />
    <br /><br />
    <asp:Calendar ID="calDate" runat="server"></asp:Calendar>
    <br />
    <asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
    <br /><br />
    <asp:Label ID="lblOutput" runat="server"></asp:Label>
</div>
```

StudentForm.aspx.cs mein class ke andar paste karo

```
protected void btnSubmit_Click(object sender, EventArgs e)
{
    string gender = "";
    if (rbMale.Checked) gender = "Male";
    else if (rbFemale.Checked) gender = "Female";

    if (chkTerms.Checked)
    {
        lblOutput.Text = "Student Name: " + txtName.Text +
                         "<br/>Gender: " + gender +
                         "<br/>Course: " + ddlCourse.SelectedValue +
                         "<br/>Date: " + calDate.SelectedDate.ToShortDateString();
    }
    else
    {
        lblOutput.Text = "Please accept Terms & Conditions";
    }
}
```

D2: Registration Form with Validation Controls

Slips: 25, 35 | Project Name: RegistrationForm | Page: Registration.aspx

Step 1: Upar wale setup se project 'RegistrationForm' banao.

Step 2: Project par right-click -> Add -> New Item -> Web Form -> Name: Registration.aspx -> Add.

Step 3: Registration.aspx ke Source tab mein form tag ke andar wala replace karo (neeche markup).

Step 4: Registration.aspx.cs kholo (aspx ke aage ka '>' arrow) aur class ke andar btnSubmit_Click method paste karo.

Step 5: Registration.aspx par right-click -> Set As Start Page -> Ctrl + F5.

Step 6: Test: khali Name chhod ke button dabao (error aayega), age 10 daalo (range error), galat email daalo, alag passwords daalo. Sab sahi daalke 'Registration Successful' dikhao.

Step 7: Agar 'UnobtrusiveValidationMode requires a ScriptResourceMapping for jquery' error aaye: Web.config kholo, <appSettings> ke andar neeche wali key line add karo.

Registration.aspx (div ki jagah)

```
<div>
    Name:
    <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
    <asp:RequiredFieldValidator ID="rfvName" runat="server" ControlToValidate="txtName"
        ErrorMessage="Name is required" ForeColor="Red" Display="Dynamic"></asp:RequiredFieldValidator>
    <br /><br />

    Age:
    <asp:TextBox ID="txtAge" runat="server"></asp:TextBox>
    <asp:RangeValidator ID="rvAge" runat="server" ControlToValidate="txtAge"
        MinimumValue="18" MaximumValue="60" Type="Integer"
        ErrorMessage="Age must be between 18 and 60" ForeColor="Red" Display="Dynamic"></asp:RangeValidator>
    <br /><br />

    Email:
    <asp:TextBox ID="txtEmail" runat="server"></asp:TextBox>
    <asp:RegularExpressionValidator ID="revEmail" runat="server" ControlToValidate="txtEmail"
        ValidationExpression="\w+([-+.']\w+)*@\w+([-.]\w+)*\.\w+([-.]\w+)*"
        ErrorMessage="Invalid email format" ForeColor="Red"
            Display="Dynamic"></asp:RegularExpressionValidator>
    <br /><br />

    Password:
    <asp:TextBox ID="txtPass" runat="server" TextMode="Password"></asp:TextBox>
    <br /><br />

    Confirm Password:
    <asp:TextBox ID="txtConfirm" runat="server" TextMode="Password"></asp:TextBox>
    <asp:CompareValidator ID="cvPass" runat="server" ControlToValidate="txtConfirm"
        ControlToCompare="txtPass" ErrorMessage="Passwords do not match"
        ForeColor="Red" Display="Dynamic"></asp:CompareValidator>
    <br /><br />

    <asp:Button ID="btnSubmit" runat="server" Text="Register" OnClick="btnSubmit_Click" />
    <br /><br />
    <asp:Label ID="lblOutput" runat="server"></asp:Label>
</div>
```

Registration.aspx.cs mein class ke andar paste karo

```
protected void btnSubmit_Click(object sender, EventArgs e)
{
    if (Page.IsValid)
    {
        lblOutput.Text = "Registration Successful";
    }
}
```

Web.config -> <appSettings> ke andar (sirf error aaye tab)

```
<add key="ValidationSettings:UnobtrusiveValidationMode" value="None" />
```

D3: AdRotator Control

Slip: 27 | Project Name: AdrotatorDemo | Page: AdrotatorDemo.aspx

Step 1: Upar wale setup se project 'AdrotatorDemo' banao.

Step 2: Project par right-click -> Add -> New Folder -> naam: Images.

Step 3: Koi bhi 2 images (jpg) copy karke Images folder mein paste karo aur naam rakho: p1.jpg aur p2.jpg.

Step 4: Project par right-click -> Add -> New Item -> 'XML File' -> Name: Ads.xml -> Add. Purana content delete karke neeche wala Ads.xml paste karo.

Step 5: Project par right-click -> Add -> New Item -> Web Form -> Name: AdrotatorDemo.aspx -> Add.

Step 6: AdrotatorDemo.aspx ke Source tab mein form tag ke andar wala replace karo (neeche markup).

Step 7: Page par right-click -> Set As Start Page -> Ctrl + F5. Page ko baar baar F5 se refresh karo, ad badalta rahega. Code-behind mein kuch nahi likhna.

Ads.xml

```
<?xml version="1.0" encoding="utf-8"?>
<Advertisements>
    <Ad>
        <ImageUrl>~/Images/p1.jpg</ImageUrl>
        <NavigateUrl>https://www.dell.com</NavigateUrl>
        <AlternateText>Dell Laptop</AlternateText>
        <Impressions>50</Impressions>
    </Ad>
    <Ad>
        <ImageUrl>~/Images/p2.jpg</ImageUrl>
        <NavigateUrl>https://www.hp.com</NavigateUrl>
        <AlternateText>HP Laptop</AlternateText>
        <Impressions>50</Impressions>
    </Ad>
</Advertisements>
```

AdrotatorDemo.aspx (div ki jagah)

```
<div>
    <asp:AdRotator ID="AdRotator1" runat="server" Width="300px" Height="200px" AdvertisementFile="~/Ads.xml"
        />
</div>
```

Exam Tips

- Sirf apni slip ka ek easy question karo. Pass hone ke liye ek kaafi hai.
- Console program mein purana code POORA delete karke hi naya paste karo, warna dono mix hoke error aata hai.
- Run karne ka shortcut: Ctrl + F5. Agar error aaye toh error ka text dhyan se padho, aksar spelling ya bracket ki galti hoti hai.
- PWA mein files wwwroot folder ke andar hi honi chahiye, aur naam bilkul same (index.html, manifest.json, service-worker.js, app.js).
- Examiner ko output dikhate waqt F12 -> Application tab mein Manifest aur Service Worker zaroor dikhao.